

Duration: 3 Hours

Note:

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- Unit ramp function $r(2t-2)$ starts at what value of t ?
A) 1
B) 2
C) -2
D) -1
- Total energy in $u(t)$ is _____
A) $\frac{1}{2}$
B) 0
C) ∞
D) 1
- The even part of the function $x(t) = \sin(t)$ is
A) $\sin(t)$
B) $-\sin(t)$
C) 0
D) $2\sin(t)$
- The period of signal $x(t) = 24 + 50 \cos(60\pi t)$ is
A) $1/30$
B) 60π
C) $1/60\pi$
D) Not periodic
- A discrete LTI system is characterized by the impulse response $h(n)=u(n)$. The output when the input is $x(n)=\{1,1,2,5\}$.
A) $\{1,3,4,5,9,9,9, \dots\}$
B) $\{1,2,4,9,9,9, \dots\}$
C) $\{1,3,4,9,9,9, \dots\}$
D) $\{2,3,4,9,9,9, \dots\}$
- The associative property of convolution sum is given by
A) $x(n)*\{h_1(n)+h_2(n)\}=x(n)*h_1(n)+x(n)*h_2(n)$
B) $x(n)*\{h_1(n)*h_2(n)\}=x(n)*h_1(n)+x(n)*h_2(n)$
C) $x(n)*\{h_1(n)*h_2(n)\}=\{x(n)*h_1(n)\}*h_2(n)$
D) $x(n)*\{h_1(n)+h_2(n)\}=x(n)+h_2(n)*x(n)+h_1(n)$
- The commutative property of convolution integral is given by
A) $x(t)*\{h_1(t)+h_2(t)\}=x(t)*h_1(t)+x(t)*h_2(t)$
B) $x(t)*\{h_1(t)*h_2(t)\}=x(t)*h_1(t)+x(t)*h_2(t)$
C) $x(t)*\{h_1(t)*h_2(t)\}=\{x(t)*h_1(t)\}*h_2(t)$
D) $x(t)*h(t)=h(t)*x(t)$
- The step response of a discrete LTI System is given by
A) $s(n) = \sum_{k=-\infty}^n h(k)$
B) $s(n) = \sum_{k=-\infty}^{\infty} h(n)$
C) $s(n) = \sum_{k=0}^{\infty} h(k)$
D) $s(n) = \sum_{k=-\infty}^0 h(k)$
- What is the outcome of a periodic convolution of signals in the frequency domain?
A) Division of signals
B) Multiplication of signals
C) Subtraction of signals
D) Addition of signals
- What is the frequency shift property of Fourier Series of the signal $x(t)$ shifted by k_0 ?
A) $X(k)=X(k+k_0)$
B) $X(k)=X(k+k_0)e^{j\omega t}$
C) $X(k)=X(k-k_0)$
D) $X(k)=X(k+k_0)e^{-j\omega t}$
- What are the conditions required for a signal to be represented as Fourier series?
A) Dirichlet's conditions
B) Gibb's phenomenon
C) Fourier conditions
D) Fourier phenomenon
- The discrete time Fourier coefficient of $\cos^2(8n\pi)$ is
A) $\pi 2(\delta(k+1) + 2\delta[k] + \delta[k-1])$
B) $14j(\delta(k+1) + 2\delta[k] + \delta[k-1])$
C) $14(\delta(k+1) + 2\delta[k] + \delta[k-1])$
D) $\pi 4(\delta(k+1) + 2\delta[k] + \delta[k-1])$

13. The frequency shift property of Fourier Transform is given by
 A) $e^{j\beta t}x(t) \leftrightarrow X(j(\omega+\beta))$ B) $e^{j\beta t}x(t) \leftrightarrow X(j(\omega-\beta))$
 C) $e^{-j\beta t}x(t) \leftrightarrow X(j(\omega-\beta))$ D) $e^{-j\beta t}x(t) \leftrightarrow X(j(\omega+\beta))$
14. The frequency differentiation property of DTFT is given by
 A) $-jx(n) \leftrightarrow \frac{d}{d\Omega} X(e^{j\Omega})$ B) $-jnx(n) \leftrightarrow \frac{d}{d\Omega} X(e^{j\Omega})$
 C) $-nx(n) \leftrightarrow \frac{d}{d\Omega} X(e^{j\Omega})$ D) $x(n) \leftrightarrow \frac{d}{d\Omega} X(e^{j\Omega})$
15. DTFT is used for the analysis of discrete time signals which are
 A) Periodic B) Non periodic
 C) Causal D) Non-Causal
16. If $x[n]$ is a signal whose DTFT is $X(\Omega)$, then if $y[n]=x[n-n_0]$ is a signal, its DTFT is
 A) $e^{j\Omega n_0}X(\Omega)$ B) $e^{-j\Omega n_0}X(\Omega)$
 C) $X(n-n_0)$ D) $X(\Omega-\Omega_0)$
17. For a stable system, in z domain the following is true.
 A) All the zeros should lie within the unit circle. B) All the zeros should lie within the infinite circle.
 C) All the poles should lie within the unit circle. D) All the poles and zeros should lie within the unit circle
18. The unilateral Z transform of a unit step signal is
 A) 1 B) $z/(z+1)$
 C) 0 D) $z/(z-1)$
19. For the Region of Convergence of a Z transform the following is true.
 A) $X(z) = \sum_{n=-\infty}^{\infty} |x(n)z^{-n}| < 0$ B) $X(z) = \sum_{n=-\infty}^{\infty} |x(n)z^{-n}| < \infty$
 C) $X(z) = \sum_{n=-\infty}^{\infty} |x(n)z^{-n}| = 0$ D) $X(z) = \sum_{n=-\infty}^{\infty} |x(n)z^{-n}| = \infty$
20. The linearity property of a Z transform is
 A) $ax(n)+by(n) \leftrightarrow aX(z)+bY(z)$ B) $ax(n)+by(n) \leftrightarrow aX(z)-bY(z)$
 C) $ax(n)+by(n) \leftrightarrow aX(z)bY(z)$ D) $ax(n)by(n) \leftrightarrow aX(z)+bY(z)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

- 1 a) i) Sketch the signal $x(t) = r(t+1) - r(t) + r(t-1)$

Marks BT* CO* PO*

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- ii) Sketch the signal $y(t)$, $x(t)$ is shown in Fig (1a)
 $y(t) = [x(t) + x(2-t)]u(1-t)$

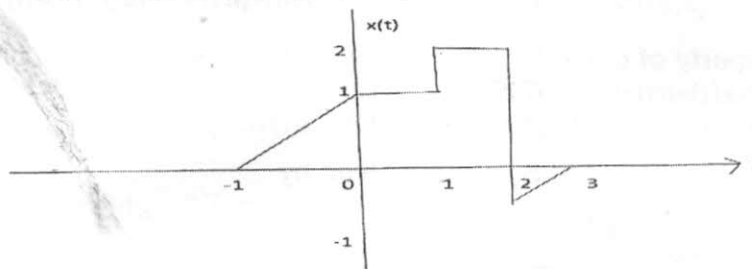


Figure (1a)

- b) Find and plot the even and odd components of the discrete-time signal.

$$x(n) = 3u(n-2)$$

- c) Determine whether the signal is periodic or not. If periodic, find its fundamental period.

$$x(t) = \cos(5\pi t) + \sin(6\pi t)$$

- d) Find the discrete-time convolution sum given below.

$$y(n) = \beta^n u(n) * \alpha^n u(n) \quad ; |\beta| < 1 \text{ \& } |\alpha| < 1$$

2 L*3 1 1

4 L3 1 1

4 L3 1 1

4 L3 2 1

2. a) For the continuous -time signal $x(t)$ shown in Figure (2a) Sketch $y(t) = x(-2t - 1)$

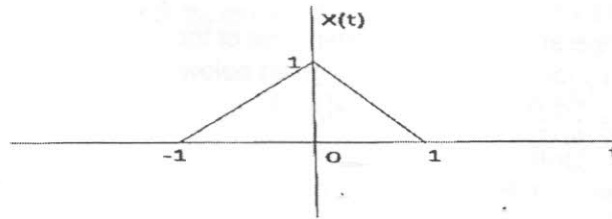


Figure (2a)

- | | | | | |
|--|---|----|---|---|
| b) Find the even and odd components of the continuous-time signals. | 4 | L3 | 1 | 1 |
| $x(t) = \cos(t) + \sin(t) + \sin(t) \cos(t)$ | 4 | L3 | 1 | 1 |
| c) Check whether the following discrete-time signals are energy or power signals. | 4 | L3 | 1 | 1 |
| $x(n) = A$ | 4 | L3 | 1 | 1 |
| d) For the impulse response of continuous-time LTI system $h(t)$ determine whether the system is memoryless, causal and stable. Justify your answer. | 4 | L3 | 2 | 1 |
| $h(t) = e^t u(t+100)$ | 4 | L3 | 2 | 1 |

3. a) A discrete -time signal

$$x(n) = \{2, 0, 2, 0, 4\}$$

Sketch i) $x(-n)$ ii) $x(2-n)$ iii) $x(-n-2)$, Also write a functional representation for each of them in terms of impulse function.

- | | | | | |
|---|---|----|---|---|
| b) Determine whether the following continuous -time signal is energy signal, power signal or neither. | 4 | L3 | 1 | 1 |
| $x(t) = A \cos(\omega_0 t + \phi)$ | 4 | L3 | 1 | 1 |
| c) Evaluate the step response for the LTI system represented by the impulse response. | 2 | L3 | 2 | 1 |
| $h(t) = \frac{1}{4}[u(t) - u(t-4)]$ | 6 | L3 | 2 | 1 |
| d) Evaluate the continuous-time convolution integral given: | 6 | L3 | 2 | 1 |
| $y(t) = e^{-2t} u(t) * u(t+2)$ | 6 | L3 | 2 | 1 |

Unit - II

- | | | | | |
|--|---|----|---|---|
| 4. a) Evaluate the DTFS representation for the signal $x(n) = \{0, 1, 2, 3\}$ where $x(n)$ is a periodic signal and sketch the spectra. Also verify Parseval's identity. | 8 | L3 | 3 | 1 |
| b) Find the Fourier transform of the signal $x(t)$ using appropriate properties. | 8 | L3 | 4 | 1 |
| $x(t) = \frac{d}{dt}[te^{-2t} \sin(t)u(t)]$ | 8 | L3 | 4 | 1 |
| 5. a) Find the Fourier series coefficients for the periodic signal $x(t)$ shown in Figure 5(a). | 8 | L3 | 3 | 1 |

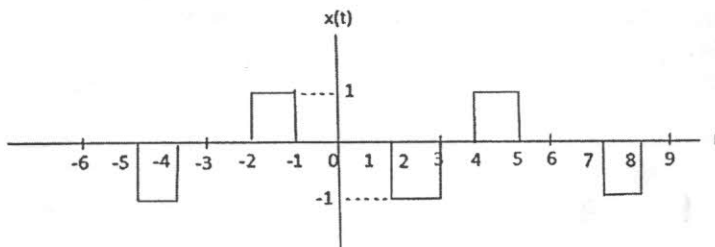


Figure 5(a)

- | | | | | |
|---|---|----|---|---|
| b) Evaluate the DTFT of the signal, | 8 | L3 | 4 | 1 |
| $x(n) = \delta(6-3n)$. Sketch its magnitude and phase spectra. | 8 | L3 | 4 | 1 |

6. a) Find the time domain signal corresponding to the DTFS coefficients $X(k)$ and sketch it.

$$X(k) = \cos\left(\frac{6\pi}{17}k\right)$$

8 L3 3 1

- b) Obtain the frequency response and the impulse response of the system having the output $y(n)$ for the input $x(n)$ as given below

$$x(n) = \left(\frac{1}{2}\right)^n u(n); y(n) = \frac{1}{4} \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n)$$

8 L3 4 1

Unit - III

7. a) Using appropriate properties, find the Z-transform of,

$$x(n) = n^2 \left(\frac{1}{2}\right)^n u(n-3)$$

5 L3 5 1

- b) Find the Z-transform of the signal

$$x(n) = 3(2)^n u(-n) \text{ using the appropriate properties.}$$

5 L3 5 1

- c) List the properties of ROC.

6 L3 5 1

8. a) Find the inverse Z-transform of,

$$X(z) = \frac{z^2 - 3z}{z^2 + \frac{3}{2}z - 1}; \text{ ROC: } \frac{1}{2} < |z| < 2$$

8 L3 5 1

- b) Determine the Z-transform of the following signals

$$x(n) = \alpha^n u(n)$$

4 L3 5 1

- c) Determine the Z-transform for the signal,

$$x(n) = \{1, 2, 5, 7, 0, 1\}$$

4 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
